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[Total No. of Questions - 9] [Total No. of Printed Pages - 3]

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MCA-6103 (Computer Organization and Architecture)

MCA 1st (NEP)

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all selecting one question each from unit I, II, III and IV. Unit V (Question No. 9) is compulsory.

UNIT - I

1. (a) Explain the process of binary subtraction using 1's and 2's complement. (6)
- (b) Describe the floating point addition algorithm with example. (6)

OR

2. (a) Discuss the Error detection codes and their importance in computer system. (6)
- (b) Perform the following conversions:
 - (i) $(101110.101)_2 \rightarrow \text{Decimal}$
 - (ii) $(A2F)_{16} \rightarrow \text{Binary}$
 - (iii) $(735)_{10} \rightarrow \text{Decimal}$ (3×2=6)

UNIT - II

3. (a) Explain the design and working of 4 bit binary adder-subtractor circuit. (6)
- (b) what is map simplification? Simplify the boolean function $F(A,B,C,D) = \sum(0,2,5,7,8,10,13,15)$ using k-map. (6)

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OR

4. (a) Describe Arithmetic micro-operations in register transfer. (6)
- (b) Explain the working of JK flip flop with truth table and excitation table. (6)

UNIT - III

5. (a) Draw and explain the general register organization of a CPU. (6)
- (b) What are different addressing modes? Explain with examples. (6)

OR

6. (a) Explain the working of micro-program sequencer with block diagram. (6)
- (b) Write a short notes on:
 - (i) Instruction cycle
 - (ii) Stack based organization (2×3=6)

UNIT - IV

7. (a) Explain the priority interrupt system with a block diagram. (6)
- (b) Explain the working of DMA with a block diagram. (6)

OR

8. (a) Discuss different level of memory hierarchy with neat diagram. (6)
- (b) Explain the concept of virtual memory and its implementation. (6)

[P.T.O.]

UNIT - V (Compulsory)

9. Write a short note on:

- (i) Define data types with examples.
- (ii) What are error detection codes? Give one example.
- (iii) Differentiate between control memory and main memory.
- (iv) What is virtual memory.
- (v) What is difference between serial and parallel communication.
- (vi) What is difference between DMA and interrupt?

(6×2=12)